

What an agent is made of

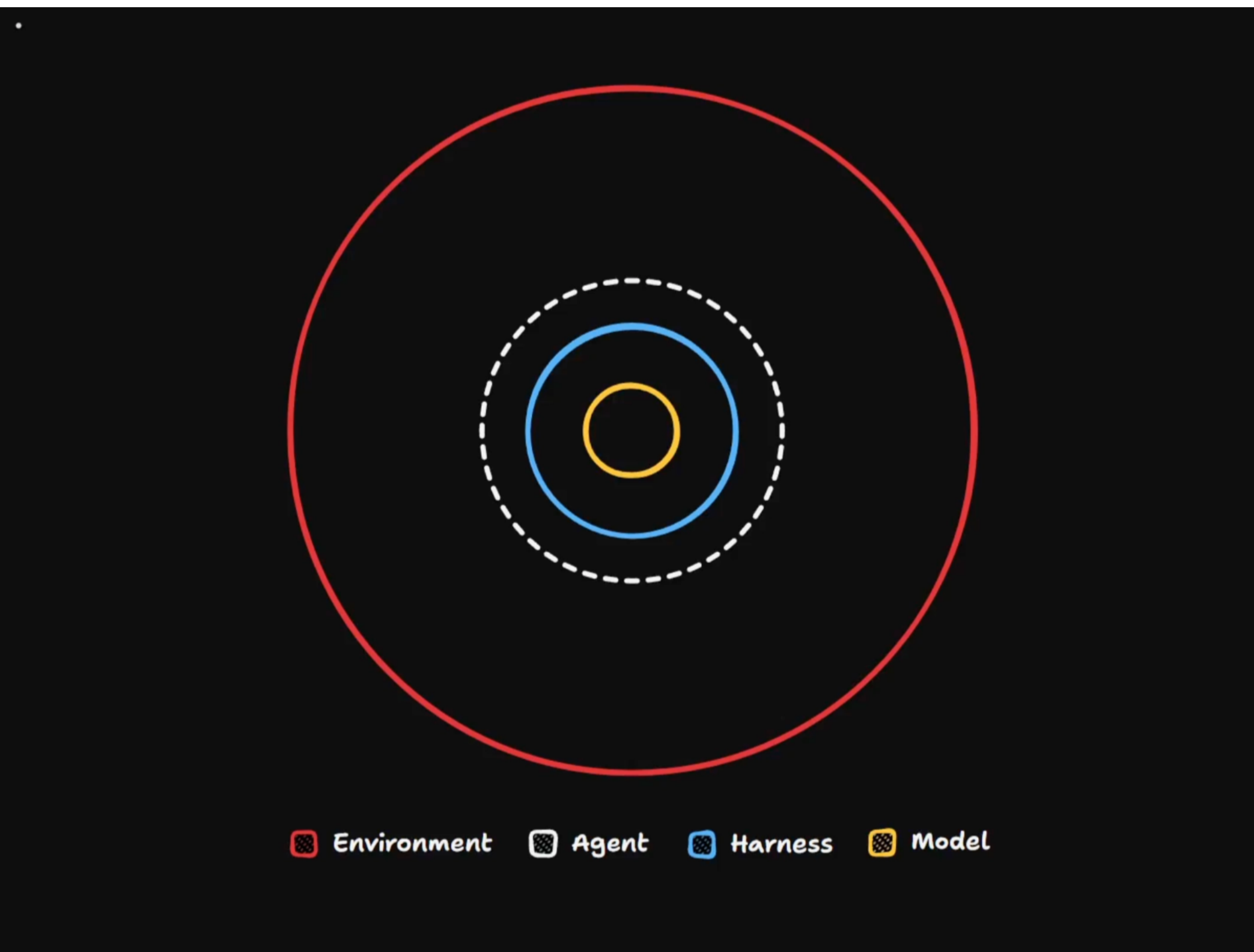
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What an Agent Is Made Of

Read this before [Lecture 1](#), alongside the first two sections of the [AI Coding Dictionary](#). Ten minutes.

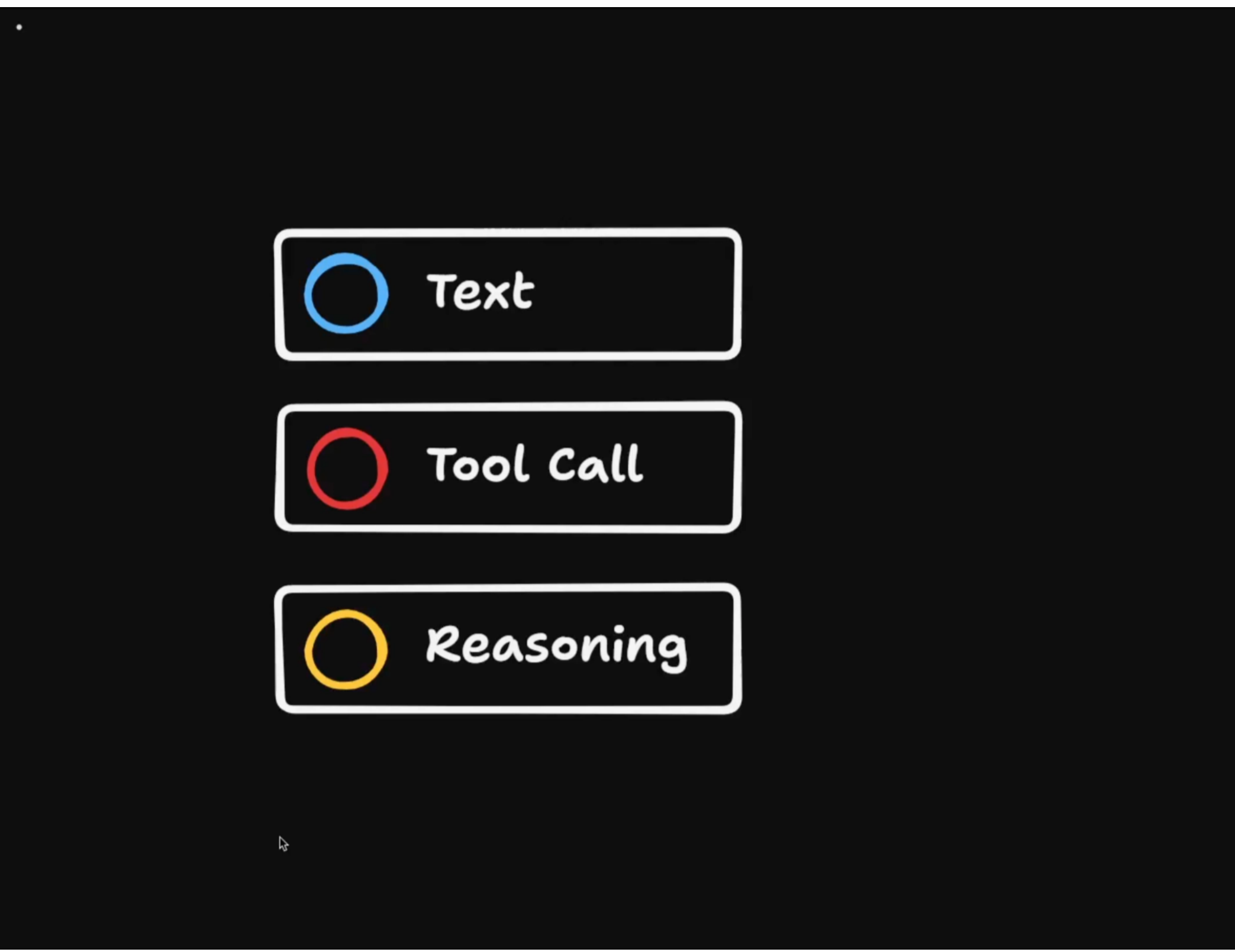
Five pictures behind the vocabulary the lectures use without stopping to define. None of it is graded directly, and none of it is hard. It is here so that when a lecture says *harness*, or *turn*, or *contextual knowledge*, the word already means something to you and the room does not have to stop.

The layers, outside in



The **model** is the thing that predicts tokens; it has no hands. The **harness** wraps it in a loop and a set of tools — Claude Code, Cursor, Codex are harnesses. Together they are the **agent**. And the agent runs inside an **environment**: a repository, a shell, a filesystem, credentials. Almost nothing you will do in this course is done to the model. You change the harness and you change the environment. That is the reason "*which model should I use*" is the least interesting question in the room — and why it gets two minutes of Lecture 1 and a [document of its own](#).

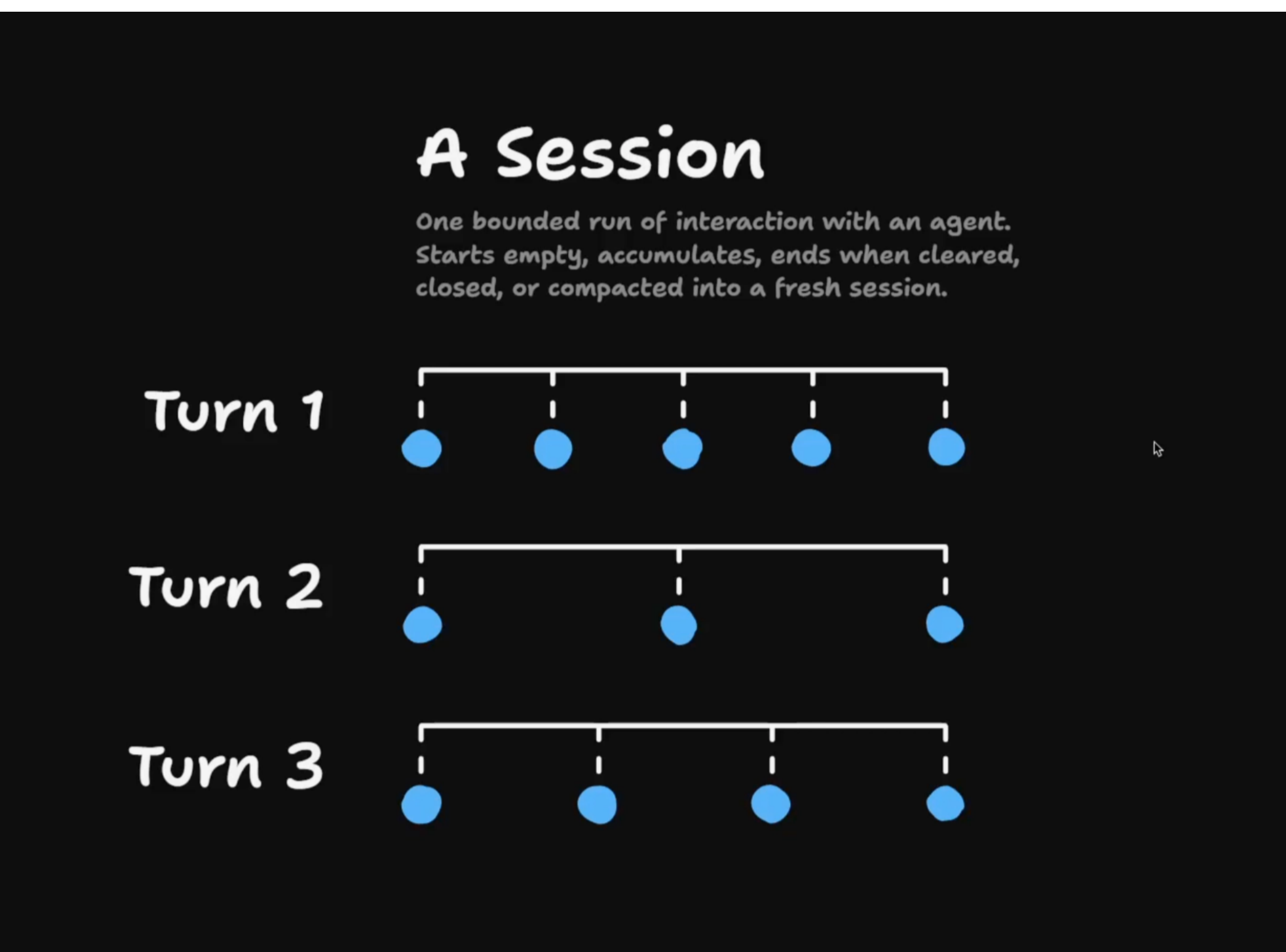
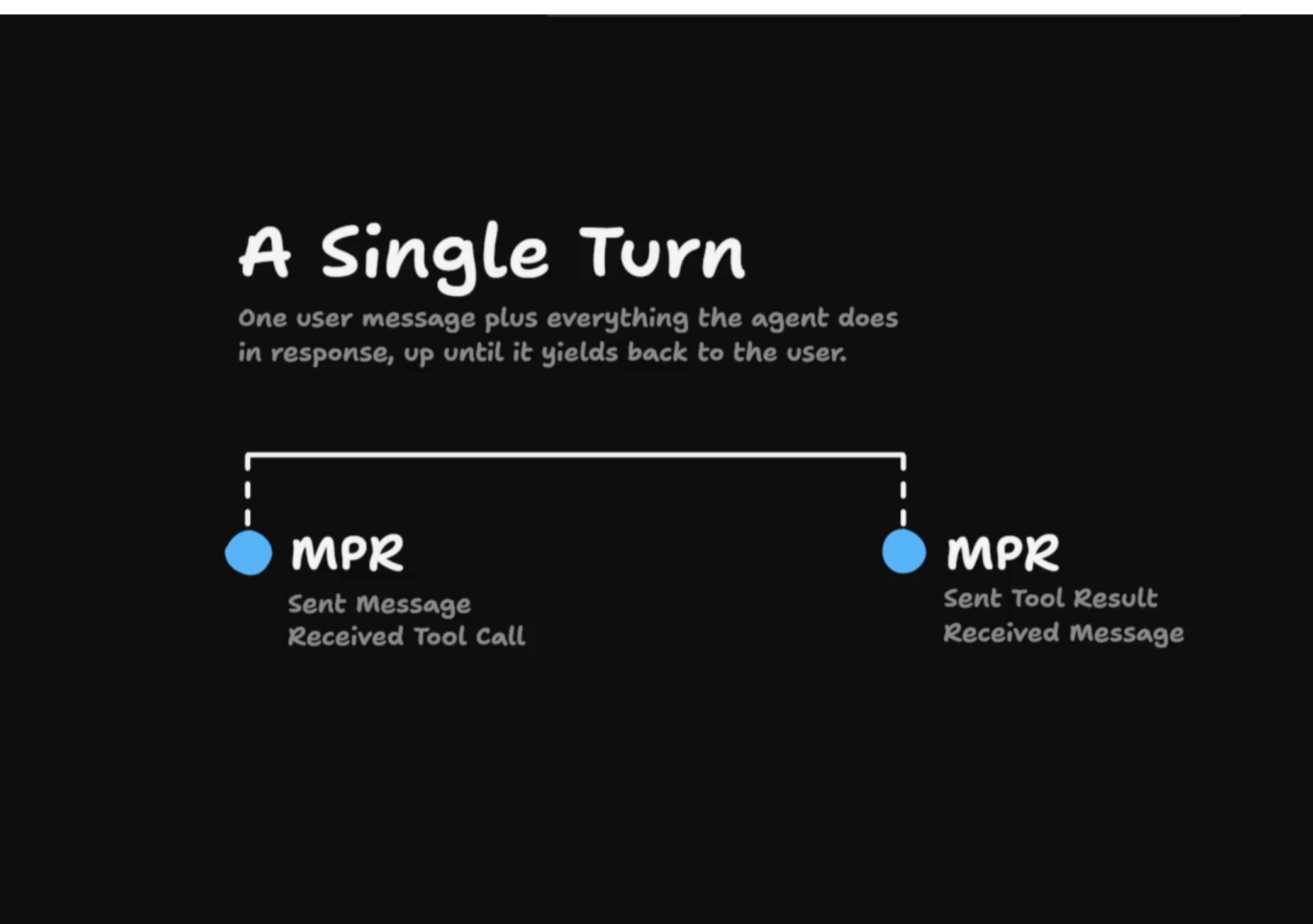
What the model can actually emit



Three things and no others. Everything an agent *does* is a tool call, which means the harness's tool list is the exact boundary of what it can do. This is also the setup for Lecture 2: a hook intercepts a tool call, which is why enforcement works without the model's cooperation.

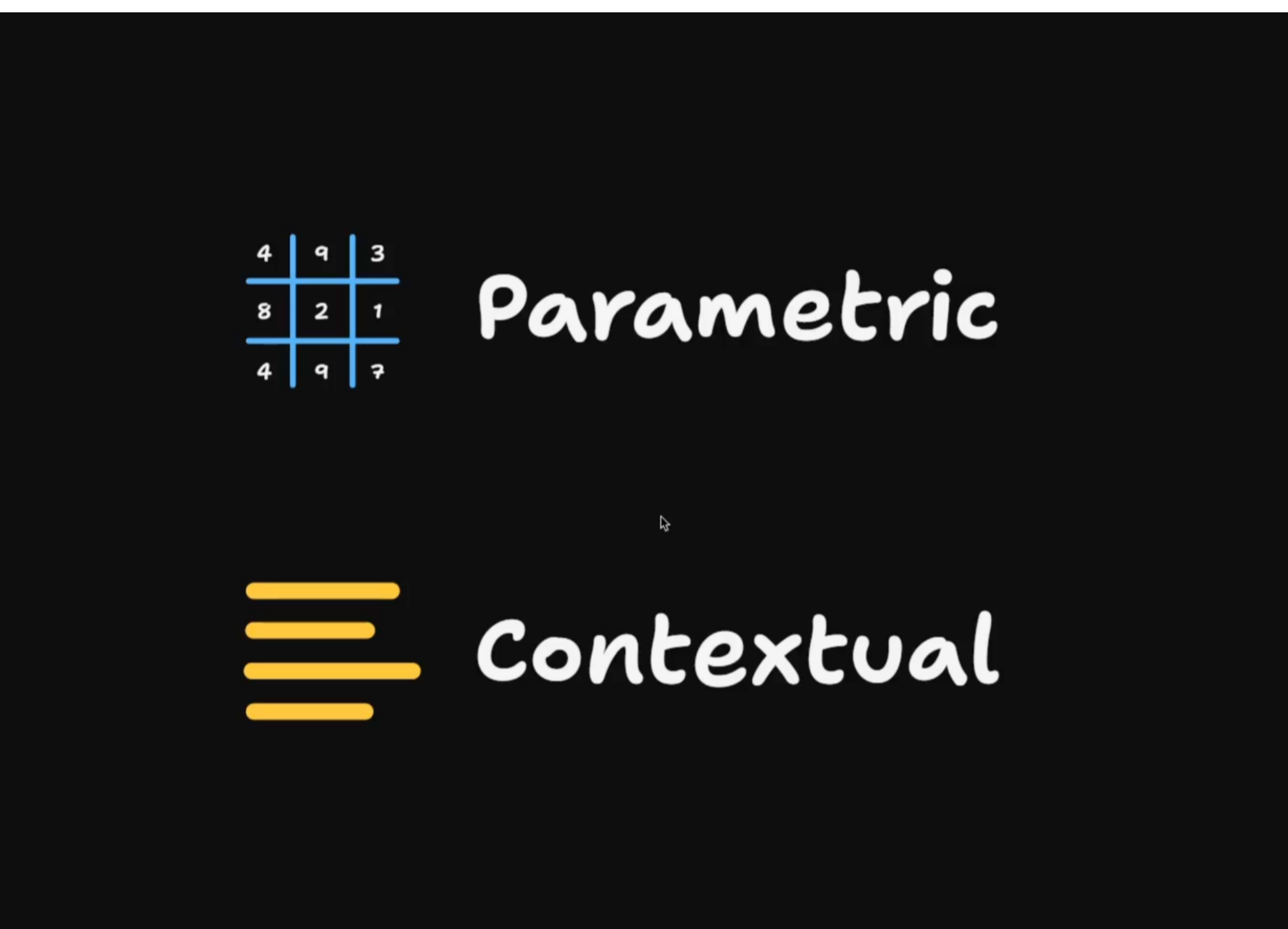
Request, turn, session

Three words students use interchangeably and should not.



One **request** is one round trip to the model. One **turn** is your message plus everything the agent does before it hands control back — often a dozen requests. One **session** starts empty, accumulates every turn, and ends when it is cleared, closed, or compacted. The unit that fills up is the session; the unit you are billed for is the request; the unit you experience is the turn. Confusing them is why "I only sent three messages" and "you spent 200k tokens" are both true.

Parametric and contextual knowledge



Parametric knowledge is baked into the weights: how Python works, what a left join is. **Contextual** knowledge is what you put in the window: your schema, your conventions, this ticket. The agent cannot tell you which is which, and it is equally confident about both — that is the mechanism behind most hallucinations you will hit. The practical rule: anything specific to your project, your company, or the last eighteen months has to be contextual, and if you did not put it there, it was guessed.